Chapter 2 / Random Signals

Issued: 13 October 2026 Due: **27 October 2026**

Name: _____ Student ID: _____

Exercise 2.3

Required

Textbook p. 61

A coin is tossed, and a random process is defined by

$$X(t) = \begin{cases} \cos \pi t, & \text{if heads occurs,} \\ 2t, & \text{if tails occurs.} \end{cases}$$

Heads and tails are equally likely. Find:

(1) The one-dimensional distribution functions $F_X(x, 1/2)$ and $F_X(x, 1)$ of $X(t)$;

(2) The two-dimensional distribution function $F_X(x_1, x_2; 1/2, 1)$ of $X(t)$;

(3) Sketch the graphs of the above distribution functions.

Answer / Working

Chapter 2 / Random Signals

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Exercise 2.13

Required

Textbook p. 62

Consider the sinusoidal voltage signal

$$X(t) = A\cos(\omega t + \Theta),$$

where ω is a constant, $A \sim U(-1, +1)$, and $\Theta \sim U(-\pi, +\pi)$. The random variables A and Θ are independent.

If this voltage is applied across a parallel RC circuit, find the total current signal and its mean-square value.

Answer / Working

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Exercise 2.16

Required

Textbook p. 62

Let the random process be

$$X(t) = A\cos\omega t + B\sin\omega t,$$

where ω is a constant and A and B are independent Gaussian random variables. Given that

$$E[A] = E[B] = 0, \qquad E[A^2] = E[B^2] = \sigma^2,$$

find the one-dimensional and two-dimensional probability density functions of $X(t)$.

Answer / Working

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Exercise 2.19

Required

Textbook p. 62

Let the random vector $(X, Y) \sim N(\boldsymbol{\mu}, \boldsymbol{C})$, where

$$\boldsymbol{\mu} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \qquad \boldsymbol{C} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}.$$

Find the joint probability density function of (X, Y) and its characteristic function $\phi_{XY}(u, v)$.

Answer / Working

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Exercise 2.1

Optional

Textbook p. 61

Consider the sinusoidal random signal $X(t) = V\cos\omega t$, where $0 \le t < \infty$, ω is a constant, and V is uniformly distributed on $[0, 1]$.

(1) Sketch two sample functions of this process.

(2) Determine the probability density function of the random variable $X(t_i)$ at each of the times

$$t_i = 0, \quad \frac{\pi}{4\omega}, \quad \frac{3\pi}{4\omega}, \quad \frac{\pi}{\omega}.$$

(3) Find the probability density function of $X(t_0)$ when $t_0 = \pi/(2\omega)$.

Answer / Working

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Exercise 2.4

Optional

Textbook p. 61

Suppose that the binary data sequence $\{B(n), n = 1, 2, 3, \ldots\}$ is a Bernoulli random sequence. Each bit corresponds to a random variable $B(n)$, with

$$P[B(n) = 0] = 0.2, \qquad P[B(n) = 1] = 0.8.$$

(1) What is the probability that four consecutive bits form the word $\{1011\}$?

(2) What is the mean word formed by four consecutive bits?

(3) Which word formed by four consecutive bits has the highest probability?

Answer / Working

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Exercise 2.7

Optional

Textbook p. 61–62

The position of a moving particle is described by the straight-line process

$$X(t) = Vt + X_0,$$

where $V \sim N(1, 1)$ and $X_0 \sim N(0, 2)$ are independent.

Find the one-dimensional probability density function, mean, and variance of the random variable $X(t)$ at time t.

Answer / Working

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Exercise 2.8

Optional

Textbook p. 62

Suppose that the Bernoulli sequence $\{I_n, n = 1, 2, \ldots\}$ takes the values -1 and $+1$ with equal probabilities.

Find its one-dimensional probability density function, mean, and covariance function.

Answer / Working

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Exercise 2.11

Optional

Textbook p. 62

Consider the two random signals

$$X(t) = A\sin(\omega t + \Theta), \qquad Y(t) = B\cos(\omega t + \Theta),$$

where A and B are random variables with unspecified distributions, Θ is uniformly distributed on $[0, 2\pi]$, and ω is a constant. The random variables A, B, and Θ are pairwise statistically independent.

(1) Find the cross-correlation function $R_{XY}(t_1, t_2)$ of the two random signals.

(2) Discuss their orthogonality, uncorrelatedness, and statistical independence.

Answer / Working

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Exercise 2.18

Optional

Textbook p. 62

A Gaussian signal has mean and covariance function

$$m_X(t) = 2, \qquad C_X(t_1, t_2) = 8\cos(t_1 - t_2).$$

Write down its three-dimensional probability density function at the times

$$t_1 = 0, \qquad t_2 = 0.5, \qquad t_3 = 1.$$

Answer / Working